Curriculum Vitae

Personal Information

Name: WU, Jingxian
Date of Birth: November 25, 1975
Gender: Male
Major: Electrical and Computer Engineering
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Education

2001 ~ Present **University of Missouri – Columbia** **MO, USA**
Candidate of Ph.D.
- Grade Average: 4.0/4.0

1998 ~ 2001 **Tsinghua University** **Beijing, P. R. China**
Master of Science, Jan. 2001
- Grade Average: 88.6/100

1994 ~ 1998 **Beijing University of Aero. & Astro.** **Beijing, P. R. China**
Bachelor of Engineering, July 1998
- Grade Average: 89.0/100

Recent Research Experiences

- **Channel estimation for TDMA-based wireless communication systems**
  This part of research focuses on the estimation of the time-varying frequency-selective fading channels in TDMA-based wireless communication systems. A polynomial interpolation based estimation algorithm with off-line training method is proposed. The new algorithm successfully balances the computational cost of the system and the estimation accuracy, and simulation results show that the performance of the algorithm is close to that of the perfect channel estimation.
- **Discrete-time channel modeling for MIMO wireless communication systems**
  A discrete-time channel model is proposed for multiple-input multiple-output (MIMO) wireless communication systems with time-varying and frequency-selective fading channels. The newly proposed discrete-time channel model can greatly reduce the computational complexity of MIMO system simulations, and its accuracy is rigorously verified through theoretical analyses and simulations.
- **Multi-user channel estimation for broadband CDMA communication systems**
  Based on the statistical properties of the discrete-time MIMO channel model, an MMSE-based multi-user channel estimation algorithm is proposed for broadband CDMA communication systems. The algorithm utilizes the correlation information among the discrete-time channel coefficients, which is overlooked by most of the previous works.

- **Frequency-domain equalization with space-time trellis coding**
  Frequency-domain equalization (FDE) is highly computationally efficient compared to conventional time-domain equalizers, and it is usually used with diversity techniques due to its linearity. The current phase of the research is focusing on the combined frequency-domain decoding and equalization for systems with space-time trellis coding, and the expected result is to achieve efficiency and accuracy simultaneously.

Previous Research Experiences

- HFC return path noise suppression
- MAC layer resource allocation for HFC networks
- Design and development of user end devices of HFC network
- Software design for embedded Real Time Operating Systems (PDA, Mobile Phone)
- Development of Visual SAR (a UNIX-based Synthetic Aperture Radar image processing software)

Work Experiences

- Reviewer for IEEE Journal on Selected Areas of Communications
- Reviewer for IEEE Transactions on Circuits and Systems - I
- Reviewer for IEEE Transactions on Wireless Communications
- Reviewer for International Journal on Wireless Information Networks
- Reviewer for IEEE Wireless Communications and Networking Conference (2003)
- Volunteer for Special Olympics of Missouri (2002)
- Internship engineer in the Beijing IStar Company (1999)

Academic Honors

- Research Assistantship (2001, 2002, MU)
- Guanghua Fellowship (1999, Tsinghua University**)**
- Internship Scholarship, First Prize (1999, Tsinghua University)
- Privilege to be enrolled in the master program of Tsinghua University with entrance examination waived (1998, Tsinghua University)
- The People's Scholarship, First Prize (1995, 1996, 1997, BUAA)
- Second Awards of the Advanced Mathematical Competition (1996, BUAA)
- Lizheng Fellowship (1994, BUAA)

Skills

- Communication system simulation with Matlab
- Proficient with computer programming in C/C++ on both Windows and UNIX platform
- Software design and hardware implementation of communication systems.
- Familiar with PCB design tools, such as PowerPCB, Protel, ORCAD

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